Recall: n-th order linear ODE $a_n(x)y^{(n)} + \cdots + a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$

- General solution: $y(x) = y_C + y_P$
- y_C via differential operators on associated homo. ODE (ahODE)
  - $a_n(x), \ldots, a_0(x)$ must be constants
- y_P via MUC
  - $a_n(x), \ldots, a_0(x)$ must be constants
  - $g(x)$ must be a finite sum or a finite product of certain forms
  - ODE need not be in standard form
  - y_P follows form of $g(x)$, but multiplied by x until the form no longer appears in y_C
- y_P via MVP
  - ODE must be in standard form
  - $y_P = u_1 y_{C,1} + u_2 y_{C,2} + \cdots + u_n y_{C,n}$
  - Solve for $u_1', \ldots, u_n'$ from a system of n equations

Euler-Cauchy (EuChy) equations

$a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \cdots + a_2 x^2 y'' + a_1 x^1 y' + a_0 y = g(x)$ $a_n, \ldots, a_0$ are constants

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Strategy for solving

- General solution: $y(x) = y_C + y_P$
- Sove for y_C from ahODE
  - Set $g(x) = 0$
  - Substitute $y = x^m$ to ahODE
  - Form polynomial in m (aka, auxiliary eqn)
  - Solve for roots $m_1, \ldots, m_n$ of auxiliary eqn
    - distinct real roots: $y_C = C_1 x^{m_1} + C_2 x^{m_2} + \cdots$
    - for each repeating root: multiply by $\ln x$
    - complex roots $\alpha \pm j\beta$: $x^\alpha (D \cos(\beta \ln x) + E \sin(\beta \ln x))$
- Sove for y_P via MVP

Example 1 $2x^3y''' + 8x^2y'' + 2xy' - 2y = x + 1$

Consider ahODE: $2x^3y''' + 8x^2y'' + 2xy' - 2y = 0$

Apply $y = x^m$: $y' = mx^{m-1}$ $y'' = m(m-1)x^{m-2}$ $y''' = m(m-1)(m-2)x^{m-3}$

Auxiliary eqn: $(2m^3 + 2m^2 - 2m - 2)x^m = 0 \quad \Rightarrow \quad m = 1, -1, -1$

$$y_C = C_1x + C_2x^{-1} + C_3x^{-1}\ln x$$

Example 1 $2x^3y''' + 8x^2y'' + 2xy' - 2y = x + 1$

Standard form: $y''' + \frac{4}{x}y'' + \frac{1}{x^2}y' - \frac{1}{x^3}y = \frac{x+1}{2x^3} = \frac{1}{2}x^{-2} - \frac{1}{2}x^{-3}$

$y_C = C_1x + C_2x^{-1} + C_3x^{-1}\ln x$

$y_P = u_1x + u_2x^{-1} + u_3x^{-1}\ln x$

Solve for $u_1', \ldots, u_3'$:

$$\begin{bmatrix} x & x^{-1} & x^{-1}\ln x \\ 1 & -x^{-2} & x^{-2}(1-\ln x) \\ 0 & 2x^{-3} & x^{-3}(2\ln x - 3) \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} = f(x) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$W = \det\begin{pmatrix} x & x^{-1} & x^{-1}\ln x \\ 1 & -x^{-2} & x^{-2}(1-\ln x) \\ 0 & 2x^{-3} & x^{-3}(2\ln x - 3) \end{pmatrix} = 4x^{-4}$$

$$W_1 = \det\begin{pmatrix} 0 & x^{-1} & x^{-1}\ln x \\ 0 & -x^{-2} & x^{-2}(1-\ln x) \\ 1 & 2x^{-3} & x^{-3}(2\ln x - 3) \end{pmatrix} = x^{-3}$$

$$W_2 = \det\begin{pmatrix} x & 0 & x^{-1}\ln x \\ 1 & 0 & x^{-2}(1-\ln x) \\ 0 & 1 & x^{-3}(2\ln x - 3) \end{pmatrix} = x^{-1}(2\ln x - 1)$$

$$W_3 = \det\begin{pmatrix} x & x^{-1} & 0 \\ 1 & -x^{-2} & 0 \\ 0 & 2x^{-3} & 1 \end{pmatrix} = -2x^{-1}$$

$$u_1' = \frac{W_1}{W}f(x) = \frac{x^{-1}}{8} + \frac{x^{-2}}{8}$$

$$u_2' = \frac{W_2}{W}f(x) = \frac{1}{8}(2x\ln x - x + 2)$$

$$u_3' = \frac{W_3}{W}f(x) = -\frac{x}{4} - \frac{1}{4}$$

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General solution: $$y(x) = C_1x + C_2x^{-1} + C_3x^{-1}\ln x + \frac{x\ln x}{8} - \frac{1}{2}$$

Example 2 $2(3+x)^2y'' + 4(3+x)y' + 5y = x - 7$

Let $t = 3 + x$: $2t^2y'' + 4ty' + 5y = t - 10$

Consider ahODE; apply $y = t^m$: $y' = mt^{m-1}$ $y'' = m(m-1)t^{m-2}$

Auxiliary eqn: $(2m^2 + 2m + 5)t^m = 0 \quad \Rightarrow \quad m = -0.5 \pm j1.5$

$$y_C = C_1 t^{-0.5}\cos(1.5\ln t) + C_2 t^{-0.5}\sin(1.5\ln t)$$

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Standard form: $y'' + 2t^{-1}y'' + 2.5t^{-2}y = \frac{t-10}{2t^2} = \frac{1}{2}t^{-1} - 5t^{-3}$

Solve for $u_1', \ldots, u_3'$:

$$\begin{bmatrix} y_{C,1} & y_{C,2} \\ y_{C,1}' & y_{C,2}' \end{bmatrix}\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(t) \end{bmatrix} = f(t)\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Some nerd things later...

$$y_P = \frac{t}{9} - 2$$

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General solution: $y(t) = C_1 t^{-0.5}\cos(1.5\ln t) + C_2 t^{-0.5}\sin(1.5\ln t) + \frac{t}{9} - 2$

Getting back with x:

$$y(x) = C_1(3+x)^{-0.5}\cos(1.5\ln(3+x)) + C_2(3+x)^{-0.5}\sin(1.5\ln(3+x)) + \frac{3+x}{9} - 2$$

Example 3 $2(3-2x)^2y'' + 4(3-2x)y' + 5y = x - 7$

Pro (sub)stitution: $t = 3 - 2x$

"Reparameterization" trick: $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

Putting our maggots4brainz to work: $-4t^2y'' - 8ty' + 5y = \frac{11 - t}{2}$